

Documentation Report

KRITI 2026 — DF-Only (2010–2020) Dual-Engine Long-Only Trading Strategy

Largecap: CPR + XGB — Small/Mid: MA + XGB + Breakeven Ratchet + Tight ATR Trail

Keywords:

1. Introduction

This report documents a **DF-only** (dataset-only) systematic trading strategy designed for KRITI 2026 constraints. The strategy is **long-only**, uses **integer shares**, applies **strict T+1 execution** using typical price, and models **transaction costs of 0.268% per side**.

The framework uses a **dual-engine architecture**:

- **Largecap engine:** Monthly Central Pivot Range (CPR) proximity + regime-aware XGBoost regression score.
- **Small/Mid engine:** Moving-average proximity + regime-aware XGBoost regression score, augmented with a **breakeven ratchet** and **tight ATR trailing stop**.

Both engines share a single cash pool and are executed within a unified portfolio backtester that enforces KRITI constraints and eliminates look-ahead bias.

2. Compliance and Trading Assumptions

2.1. Hard Constraints

The implementation enforces:

- **Long-only, no leverage, no fractional shares**
- **Initial capital:** 50,00,000
- **Max holdings:** 100 open positions
- **Min hold days:** 1 day
- **Max hold days:** 100 days
- **Min holdings target:** 1 holding whenever possible (subject to signal availability and cash constraints)

2.2. Execution Model (Strict T+1)

Signals are computed on day T using data available up to T . Trades execute on $T + 1$ at the **typical price**:

$$P_{T+1}^{exec} = \frac{O_{T+1} + H_{T+1} + L_{T+1} + C_{T+1}}{4} \quad (1)$$

2.3. Transaction Costs

A fixed cost of 0.268% is applied per side at execution price:

$$P_{T+1}^{buy} = P_{T+1}^{exec}(1 + c), \quad (2)$$

$$P_{T+1}^{sell} = P_{T+1}^{exec}(1 - c), \quad (3)$$

where $c = 0.00268$.

3. Data and Universe Construction

3.1. Dataset and Required Columns

The strategy uses only the provided DataFrame (`df`) and requires the following columns:

- `tradedate`, `open`, `high`, `low`, `close`, `traded_volume`, `fid`, `cap_classification_NSE500`

All computations use daily bars. The strategy window is restricted to:

- **Backtest / report window:** 2010-01-01 to 2020-01-01

3.2. Internal Split (DF-only)

To reduce overfitting while remaining DF-only, an internal split is used:

- **Train:** 2010–2018 (model fit + threshold calibration)
- **Validation:** 2019 (invest fraction selection)
- **Final report backtest:** 2010–2020 (equity curve and headline metrics)

3.3. Cap Bucket Routing

The strategy normalizes `cap_classification` (lowercase, removing spaces/hyphens) and routes instruments into:

- **largecap** → Largecap engine
- **smallcap/midcap/other** → Small/Mid engine (denoted `smallmid`)

4. Indicators and Feature Engineering (Causal)

All features are built per stock (grouped by `fid`) and are causal.

4.1. Average True Range (ATR)

True range is defined:

$$TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|) \quad (4)$$

ATR uses an EWMA estimate:

$$ATR_t = \text{EWMA}(TR_t; n) \quad (5)$$

The strategy uses ATR_{21} (volatility normalization and stop sizing) and ATR_{14} (SuperTrend).

4.2. SuperTrend Direction

Using $HL2_t = (H_t + L_t)/2$:

$$Upper_t = HL2_t + m \cdot ATR_{14,t}, \quad (6)$$

$$Lower_t = HL2_t - m \cdot ATR_{14,t}, \quad (7)$$

with $m = 3.0$. The indicator outputs:

$$ST_dir_t \in \{+1, -1\}$$

4.3. Causal Gaussian Smoothing and Channel Position

A causal Gaussian-like smoother is approximated by applying six EMA passes to the close:

$$G_t = \underbrace{\text{EMA}(\text{EMA}(\dots \text{EMA}(C_t) \dots))}_{6 \text{ passes}} \quad (8)$$

Gaussian channel bounds:

$$G_t^{upper} = G_t + k \cdot ATR_{21,t}, \quad G_t^{lower} = G_t - k \cdot ATR_{21,t} \quad (9)$$

with $k = 4.0$. Normalized channel position:

$$G_t^{pos} = \frac{C_t - G_t^{lower}}{G_t^{upper} - G_t^{lower}} \quad (10)$$

4.4. Log Volume

$$V_t^{log} = \log(1 + Volume_t) \quad (11)$$

4.5. Largecap Features: Monthly CPR (Previous Month)

From the previous month's $H^{m-1}, L^{m-1}, C^{m-1}$:

$$Pivot^{m-1} = \frac{H^{m-1} + L^{m-1} + C^{m-1}}{3}, \quad (12)$$

$$BC^{m-1} = \frac{H^{m-1} + L^{m-1}}{2}, \quad (13)$$

$$TC^{m-1} = 2 \cdot Pivot^{m-1} - BC^{m-1}. \quad (14)$$

Width:

$$W^{CPR} = |TC^{m-1} - BC^{m-1}|$$

Normalized distances:

$$d_t^{pivot} = \frac{C_t - Pivot^{m-1}}{W^{CPR}}, \quad d_t^{BC} = \frac{C_t - BC^{m-1}}{W^{CPR}}, \quad d_t^{TC} = \frac{C_t - TC^{m-1}}{W^{CPR}}$$

4.6. Small/Mid Features: Moving Average Distances Normalized by ATR

$$SMA7_t = \text{SMA}(C_t, 7), \quad EMA50_t = \text{EMA}(C_t, 50)$$

ATR-normalized distances:

$$d_t^{SMA7} = \frac{C_t - SMA7_t}{ATR_{21,t}}, \quad d_t^{EMA50} = \frac{C_t - EMA50_t}{ATR_{21,t}}, \quad d_t^{SMA7-EMA50} = \frac{SMA7_t - EMA50_t}{ATR_{21,t}}$$

4.7. Final Feature Sets

4.7.1. Largecap Feature Vector

$$X_t^{LC} = \left[V_t^{log}, ST_dir_t, AT R_{21,t}, d_t^{pivot}, d_t^{BC}, d_t^{TC}, G_t^{pos} \right]$$

4.7.2. Small/Mid Feature Vector

$$X_t^{SM} = \left[V_t^{log}, ST_dir_t, AT R_{21,t}, d_t^{SMA7}, d_t^{EMA50}, d_t^{SMA7-EMA50}, G_t^{pos} \right]$$

5. Regime Detection

Regime is computed **per stock** using the smoothed series G_t . In a rolling window of up to 252 points, local extrema are extracted from sign changes in ΔG_t . Using the last four extrema:

- **Uptrend:** last two maxima increase and last two minima increase
- **Downtrend:** last two maxima decrease and last two minima decrease
- **Sideways:** otherwise or insufficient extrema

$$Regime_t \in \{\text{up, down, sideways}\}$$

6. Machine Learning Model

6.1. Target Variable

The supervised label is a 21-day forward return:

$$R_{t+21} = \frac{C_{t+21}}{C_t} - 1 \quad (15)$$

This label is used only during training. Trading remains strictly causal.

6.2. Model Type and Parameters

Two separate XGBoost regressors are trained (one per engine) using the training window (2010–2018). Both use:

- $n_estimators = 600$
- $max_depth = 5$
- $learning_rate = 0.03$
- $subsample = 0.8$
- $colsample_bytree = 0.8$
- $min_child_weight = 5$
- $reg_lambda = 1.0$
- **objective:** `reg:squarederror` (default for squared error)

The model output score per instrument/day is denoted $XGy_{i,t}$.

7. Threshold Calibration (Regime-Aware)

Model scores are converted into stable discrete decisions using per-regime thresholds:

$$tBuy_r, tSell_r \quad \text{for } r \in \{\text{up}, \text{down}, \text{sideways}\}$$

7.1. Calibration Procedure

For each regime r , candidate thresholds are searched from quantiles of XGy :

- Buy candidates: 50th to 98th percentile
- Sell candidates: symmetric low tail (via $1 - q$)

The chosen thresholds maximize directional purity under minimum coverage:

- Choose $tBuy_r$ maximizing $\Pr(R_{t+21} > 0 \mid XGy \geq tBuy_r)$
- Choose $tSell_r$ maximizing $\Pr(R_{t+21} < 0 \mid XGy \leq tSell_r)$
- Enforce $tSell_r \leq tBuy_r$ for boundary consistency

If a regime has insufficient samples (e.g., < 2000), calibration falls back to the pooled distribution.

8. Portfolio Construction and Execution

8.1. Unified Portfolio

A single portfolio trades all instruments across both engines. Positions are stored as:

- `fid`, shares (integer), entry price, entry ATR, stop price, hold days

8.2. Entry Logic

On day T , for any instrument not already held:

$$\text{Enter if } XGy_{i,T} \geq tBuy_{Regime_T} \quad (16)$$

Execution occurs on day $T + 1$ at P_{T+1}^{buy} including costs.

8.3. Exit Logic

8.3.1. Signal Exit

For held instruments, sell signal is generated if:

$$XGy_{i,T} \leq tSell_{Regime_T} \quad (17)$$

Execution occurs on day $T + 1$ at P_{T+1}^{sell} net of costs, subject to minimum hold days.

8.3.2. Forced Exit

Positions are forcibly sold if:

- stop-loss is breached (using L_{T+1} compared to stop price), or
- holding days reach 100.

8.4. Stop Loss and Small/Mid Enhancements

8.4.1. Base Stop (All Buckets)

At entry:

$$SL_0 = EntryPx - 1.5 \cdot ATR_{21}^{entry} \quad (18)$$

8.4.2. Small/Mid Breakeven Ratchet and Tight ATR Trail

Small/Mid positions maintain:

- *HighestClose* since entry
- breakeven activation when:

$$HighestClose - EntryPx \geq 1.0 \cdot ATR^{entry}$$

- once active, trail stop:

$$Trail_t = HighestClose_t - 1.0 \cdot ATR_{21,t}$$

- enforce:

$$SL_t = \max(SL_t, EntryPx, Trail_t)$$

This mechanism protects small/mid winners by locking in profits and reducing drawdown contribution.

8.5. Capital Allocation and Invest Fractions

The strategy uses two invest fractions:

$$f_{LC}, f_{SM} \in [0, 1]$$

On each day, budgets are allocated from current cash:

$$Budget_T^{LC} = f_{LC} \cdot Cash_T, \quad Budget_T^{SM} = f_{SM} \cdot Cash_T$$

Within each bucket, allocation is proportional to non-negative scores:

$$w_i = \frac{\max(XGy_i, 0)}{\sum_j \max(XGy_j, 0)} \quad (19)$$

Shares purchased:

$$Shares_i = \left\lfloor \frac{w_i \cdot Budget_T}{P_{i,T+1}^{buy}} \right\rfloor$$

8.6. Invest Fraction Calibration

(f_{LC}, f_{SM}) are selected using a grid search over:

$$\{0.20, 0.35, 0.50, 0.65, 0.80, 0.95\}^2$$

The objective is to maximize validation Sharpe ratio over 2019 (if available), otherwise fallback to training Sharpe.

9. Benchmark and Reporting

9.1. DF-Only NSE500 Proxy Benchmark

A dataset-only benchmark proxy is constructed using rows with `in_NSE500=True`. Close-to-close returns are computed per stock and averaged daily (equal-weight):

$$r_t^{bench} = \frac{1}{N_t} \sum_{i \in NSE500} r_{i,t}$$

9.2. Performance Metrics

The report computes:

- Final equity and equity curve
- CAGR
- Annualized volatility
- Maximum drawdown
- Information ratio vs NSE500 proxy
- Up capture and down capture vs NSE500 proxy
- Turnover ratio: total traded value divided by average equity
- Average and maximum holdings
- Rolling active return (1Y, 3Y, 5Y) summary (average and worst)

10. Results

All results are generated by running the DF-only notebook end-to-end with the provided Parquet dataset. The final section of the code produces:

- Equity curve plot (2010–2020)
- Drawdown plot
- Holdings count plot
- Trade log head/tail for auditability

Numerical values depend on the dataset instance and are reported directly by the notebook output.

11. Conclusion

This KRITI 2026 dual-engine strategy provides:

- Causal feature computation and regime detection,
- Separate ML models tailored to largecap vs small/mid behavior,
- Robust, regime-aware thresholds to reduce churn and noise trades,
- Execution-realistic backtesting with T+1 typical price and per-side costs,

- Risk controls including ATR stops and a small/mid breakeven ratchet with tight trailing stop,
- DF-only benchmark proxy and comprehensive reporting metrics.

The architecture is intentionally modular (feature build → model fit → threshold calibration → unified backtest) to facilitate auditability and compliance.

Availability of Data and Code

Data is provided under competition guidelines. Code is contained in the DF-only dual-engine notebook implementing: data preparation, per-stock feature engineering, dual XGBoost training, regime-aware threshold calibration, unified strict-execution backtest, benchmark proxy construction, and performance reporting.

References